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Smart BI: Transforming Data into Actionable Insights using AI

Project Domain

Excel and Power BI with Copilot

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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Year - 2026

3. Problem Statement

In today's highly competitive retail environment, data is one of the most valuable business assets. Retail organizations generate vast amounts of data across sales, inventory, operations, and customer interactions. However, despite this abundance, many businesses fail to leverage it effectively due to fragmented data systems, inconsistent formats, and incomplete records.

From a business standpoint, this creates significant operational challenges. Decision-makers often rely on manual reporting processes that are time-consuming, error-prone, and lack real-time accuracy. As a result, organizations struggle to gain clear visibility into key performance indicators such as sales trends, product performance, and inventory levels.

This limited visibility directly impacts business outcomes. Retailers face issues such as overstocking or stock shortages, missed revenue opportunities, inefficient resource allocation, and an inability to respond quickly to changing customer demands. Ultimately, this leads to reduced customer satisfaction and weakened competitive positioning in the market.

The core business challenge lies in the absence of an integrated, automated system that can transform raw data into actionable insights. Without such a system, decision-making remains delayed, inconsistent, and reactive rather than proactive.

To overcome this challenge, there is a critical need for a solution that leverages Business Intelligence and AI-driven analytics to automate data processing and reporting. Such a system can enable real-time insights, improve operational efficiency, support strategic decision-making, and drive overall business growth.

4. Objectives

- To design an AI-driven BI solution tailored for retail performance data.

- To automate summarization and reporting, reducing manual effort.

- To calculate and monitor KPIs such as revenue, profit margin, monthly growth, and customer satisfaction.

- To provide category-wise and region-wise performance analysis for identifying strengths and inefficiencies.

- To deliver executive-level insights and risk indicators using AI prompts.

5. Methodology / Implementation Plan

#	Step	Description	Stakeholder Value
1	**Domain Understanding & Data Context**	Study retail-specific challenges such as seasonal demand fluctuations, inventory management, and customer satisfaction tracking. Map stakeholder needs (executives, managers, analysts).	Ensures the solution is aligned with real-world retail problems and stakeholder expectations
2	**Data Collection & Cleaning**	Gather structured retail datasets (500+ monthly records). Remove duplicates, handle missing values, standardize formats, and validate calculated fields (profit = revenue – operational cost).	Provides a reliable dataset for accurate analysis and reporting.
3	**KPI Definition & Calculation**	Define and compute KPIs: total revenue, total profit, profit margin, monthly growth rate, loss-making categories, and average customer satisfaction.	Enables management to track performance metrics consistently.
4	**Business Intelligence Analysis**	Perform category-wise and region-wise evaluations. Identify revenue drivers, cost leakages, and performance gaps.	Helps managers and executives pinpoint strengths and inefficiencies.
5	**AI-Driven Summarization & Reporting**	Use AI prompts to generate automated summaries, executive insights, and risk indicators. Deliver dashboards and decision-ready reports.	Provides faster, more accurate insights for strategic decision-making.

6. Resources and Timeline

A. Hardware & Software Requirements

1. Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)

2. Operating System: Windows 10 / 11 or macOS

3. Microsoft Excel (Microsoft 365 / Excel 2019 or later)

4. Power BI Desktop — latest version (free download from Microsoft)

5. Microsoft Copilot — via Microsoft 365 subscription

6. AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based summarization and report generation

7. Internet connection — required for Copilot and AI Chatbot access

8. Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- Dataset: Monthly business performance data (500+ records across departments and regions).
- Attributes: Month, Department, Region, Transactions, Revenue, Operational Cost, Profit, Customer Satisfaction Score, AI Insight Indicator.
- Source: Structured dataset prepared for BI analysis (can be simulated or sourced from Kaggle/UCI Repository if needed).
- Purpose: Supports KPI calculation, BI dashboards, and AI-driven summarization for retail/business intelligence context.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	KPI calculation, BI analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

- A **high-quality, reliable dataset** with cleaned and validated data, ensuring accuracy and consistency for business analysis, along with clear documentation of data preprocessing to support transparency and auditability.
- An **interactive Power BI dashboard / analytical report** that provides real-time visibility into key business metrics such as revenue, profit margins, growth trends, regional performance, and departmental efficiency—enabling faster and more informed decision-making.
- Implementation of **AI-driven insights** using tools like Microsoft Copilot or AI chatbots to automatically generate executive summaries, highlight risks, and uncover trends—reducing manual effort and improving strategic clarity.
- A **centralized project repository** on GitHub containing all datasets, reports, dashboards, and documentation—ensuring easy access, collaboration, and scalability for future business use.

- **Retail business impact** through actionable insights such as identification of high-performing and loss-making categories, optimization of inventory and sales strategies, and improved understanding of customer satisfaction—ultimately contributing to increased profitability and operational efficiency.

8. References

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